Lab 1 – GradPace Product Description

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1 Introduction

Workforce expectations are changing rapidly due to automation, artificial intelligence, and global competition. Employers are no longer looking for just a degree. They expect candidates to have career-ready skills and relevant experience. According to the World Economic Forum, 63% of employers identify skill gaps as the biggest barrier to growth between 2025 and 2030 (World Economic Forum, 2025). At the same time, 66% of companies are decreasing entry-level hiring because of AI, and 91% report that automation is changing or eliminating jobs (Laurent, 2026). These shifts are already affecting graduates, with 50% to 60% underemployed and 48% feeling underqualified for entry-level roles in their field (Cengage Group, 2025). This highlights a growing disconnect between education and workforce expectations.

Although universities provide many resources, career preparation remains fragmented and mainly student driven. Academic progress is clearly structured, but professional development is not. Existing tools only address parts of the problem. Degree tracking systems focus on coursework, job boards list opportunities, and networking platforms showcase experience, but none of them guide students on how to connect these pieces into a clear career path. As a result, students identify gaps too late and make reactive decisions.

The main issue is the lack of a unified system that actually connects academic progress, professional development, and industry expectations. Students need a platform that provides continuous guidance, identifies gaps early, and offers personalized recommendations. GradPace addresses this need by serving as a unified career readiness platform that is designed to bridge academic progress and professional development, allowing students to identify skill gaps early and take guided steps towards resolving them.

2 Product Description

(TO BE COMPLETED)

- Provide a top-level description of CS 410 product for the average reader.
- Provide a summary of the solution –- and its goals and objectives. This section should be one paragraph minimum.

2.1 Key Product Features and Capabilities

(TO BE COMPLETED)

- What does it do?
- What is significant/unique/innovative about it?
- What does it accomplish?
- Describe how this solves the problem.

2.2 Major Components (Hardware/Software)

(TO BE COMPLETED)

- Provide an overview of the hardware needed to support the solution.
- Describe how it is structured based on CS 410 MFCD.
- Define and describe the software to be developed.

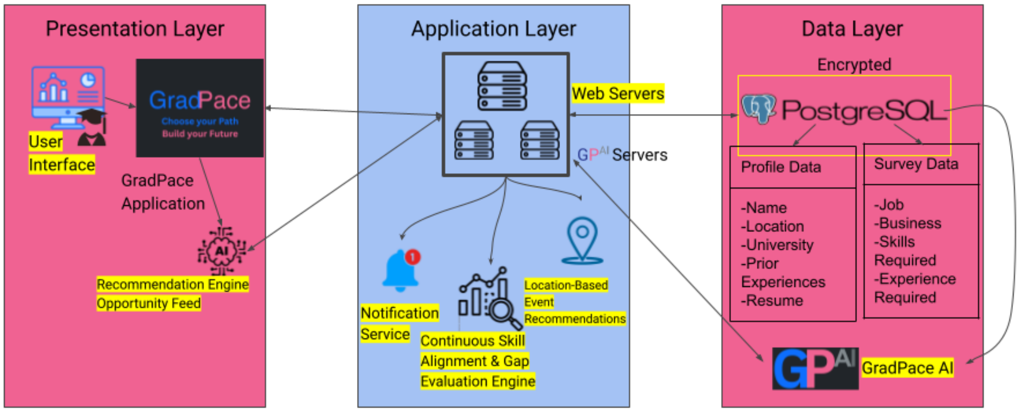

Figure 1: GradPace Major Functional Component Diagram

3 Identification of Case Study

(TO BE COMPLETED)

- For whom is this product being developed? Why?
- Identify case study group—the small group of users who will use app prototype and provide feedback.
- Who else might use this in the future?

Features	Prototype	Real World Product
Login/Authentication	Full: Core functionality and easy to implement	Secure authentication system
Account Creation/Deletion	Full: Required for all users	Full account lifecycle management
Account Subscription Management	Partial: Simulated or mocked payments	Paid subscription system
Input Academic and Professional Info	Full: Core input for system logic	User profile with professional + academic data
Personalized Roadmap Generation	Partial: Simplified logic instead of full AI	Career roadmap
View Roadmap	Full: UI-based and feasible	Interactive roadmap UI
Early-Warning Gap Detection	Partial: Rule-based instead of AI-driven	Skill gap analysis
Industry Readiness Dashboard	Partial: Basic scoring instead of dynamic benchmarking	Visual readiness scoring system
Recommendations/Guided Next Steps	Full: Static or rule-based suggestions	Guided recommendations
Notifications/Alerts	Partial: Basic notifications only	Alerts and reminders
Location-Based Event Recommendations	Partial: Static or mock event data and smaller AI model for recommendations	GPS + scraped local opportunities
Input Up-To-Date Industry Data	Eliminated: Not feasible for prototype	Continuous industry data input system
Alumni Survey	Partial: Simple form implementation	Post-graduation feedback system
Survey Results Dashboard	Partial: Basic visualization only	Aggregated alumni feedback dashboard
User Management	Partial: Limited admin interface	Admin control panel
AI Configuration/Rule Updates	Eliminated: Too complex for prototype	Dynamic AI tuning system
Data Quality Monitoring	Partial: Manual or basic validation	Automated data validation system

Table 1: Table of Comparison Between RWP and Prototype

4 Glossary

API (Application Programming Interface): A set of rules that allow different software applications to communicate and exchange data with each other.

Authentication: The process of verifying a user's identity before allowing access to a system.

Backend: The server-side part of a website or application that handles logic, databases, and data processing.

Career Readiness: A measure of how prepared a student is to enter a specific career based on skills, experience, and qualifications.

Career Roadmap: A structured, visual plan that outlines a student's progress, goals, and required steps toward a specific career path.

Cloud Computing: The use of remote servers on the internet to store data and run applications instead of a local computer.

Data Leak: The accidental or unauthorized exposure of sensitive or private information.

Database: An organized collection of data that can be stored, accessed, and managed electronically.

Encryption: A security method that converts data into a coded form so that only authorized parties can read it.

Frontend: The user-facing part of a website or application that users interact with in their browser.

Gap Detection (Early-Warning System): A feature that identifies missing skills, experiences, or qualifications needed for a target career.

Industry Readiness Benchmarking: The process of comparing a student's profile against expected skills and qualifications for specific careers.

Location-Based Event Recommendations: Suggestions for nearby internships, networking events, or career opportunities based on user location.

Professional Data Input: User-provided information such as coursework, projects, internships, and certifications used to build a profile.

Survey Results Dashboard: A feature that displays aggregated insights from alumni data to inform students and administrators.

System Administrator: A role responsible for managing the platform, maintaining data quality, and overseeing system functionality.

Token (AI): A small unit of text processed by an AI model, such as a word, part of a word, or punctuation.

User Interface (UI): The visual elements of a software application that allows users to interact with it.

Web Server: A computer system or software that stores websites and delivers web pages to users when they request them through a browser.

5 References

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